UMGS: UAV Multispectral 3D Gaussian Splatting for Spectral-Index Synthesis

ANONYMOUS AUTHOR(S)
SUBMISSION ID: TBD

UAV multispectral reconstruction is challenging because multispectral bands are weak geometry drivers. They often contain lower texture and less reliable local features than RGB, while practical UAV rigs introduce cross-FOV, resolution, and alignment differences across cameras. Direct spectral reconstruction is therefore difficult to make reliable: independently reconstructed bands can diverge in geometry and support, preventing a stable unified 3D representation for spectral-index synthesis. We introduce UMGS, a two-stage 3D Gaussian Splatting framework for UAV spectral-index synthesis. UMGS first reconstructs an RGB Gaussian anchor, then learns multispectral appearance while freezing Gaussian positions, covariances, opacity, and identity. This geometry lock lets low-texture spectral bands inherit stable 3D support from RGB while retaining band-specific appearance. We study two realizations: independent per-band transfer (UMGS-I) and joint band-bank transfer (UMGS-J). Experiments on raw UAV and aligned aerial multispectral scenes show that both UMGS variants outperform comparable multispectral Gaussian baselines overall, producing stable spectral renderings and index maps while preserving shared 3D support.

CCS Concepts: • **Computing methodologies** → **Image-based rendering**; *Reconstruction*; • **Applied computing** → *Earth and atmospheric sciences.*

Additional Key Words and Phrases: multispectral imaging, 3D Gaussian Splatting, 3D reconstruction, UAV

ACM Reference Format:
Anonymous Author(s). 2026. UMGS: UAV Multispectral 3D Gaussian Splatting for Spectral-Index Synthesis. 1, 1 (May 2026), 6 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

UAV multispectral imaging is increasingly used in precision agriculture and environmental monitoring, where decisions are made from cross-band products rather than from a single rendered view [Colomina and Molina 2014; Mathews and Jensen 2013; Turner et al. 2015; Zhang and Kovacs 2012]. A reconstruction used for spectral-index synthesis must therefore satisfy two requirements at once: accurate per-band appearance and stable geometric correspondence across bands. The latter is essential because even small support drift can bias ratio-based measurements such as NDVI, GNDVI, and NDRE [Gitelson et al. 1996; Goetz et al. 1985; Huete 1988; Manolakis et al. 2016; Rouse et al. 1974].

Raw UAV multispectral data make this difficult. Multispectral bands are weak geometry drivers: they often have lower texture and less reliable local features than RGB. Practical UAV rigs add a second source of instability because RGB and multispectral cameras can differ in field of view, resolution, and spectral response. As a result, direct spectral reconstruction or independent per-band Gaussian optimization can yield models whose geometry and support diverge across bands, preventing a stable unified 3D representation for index synthesis.

Recent radiance-field and Gaussian methods greatly improved training speed and view quality [Barron et al. 2022; Huang et al. 2024; Kerbl et al. 2023; Mildenhall et al. 2020; M"uller et al. 2022; Yu et al. 2024]. Emerging multispectral, hyperspectral, and thermal Gaussian lines [Chen et al. 2024; Lu et al. 2024a; Meyer et al. 2025; Thirgood et al. 2025] further show that spectral rendering and index synthesis are feasible. UMGS addresses the UAV-specific geometry problem: how to exploit RGB as a reliable geometric carrier while learning low-texture spectral appearance on the same 3D support.

We introduce **UMGS**, a UAV multispectral 3D Gaussian Splatting framework for spectral-index synthesis. UMGS formulates multispectral reconstruction as *RGB-anchored geometry locking*: geometry, opacity, and Gaussian identity are learned once from RGB and then frozen during G/R/RE/NIR transfer. This design turns RGB into a stable geometric carrier, while each spectral band only learns appearance on the same splat support. We evaluate two implementations of the same constraint:

- **UMGS-I**: independent per-band transfer from a shared RGB anchor.
- **UMGS-J**: joint multispectral training with band-specific appearance banks in one unified checkpoint.

Both preserve identical support by construction; they differ only in how spectral appearance parameters are organized and optimized.

For raw UAV captures, alignment quality is another bottleneck. Cross-band FOV and resolution mismatch can make naive baselines unreliable, especially for NIR. We therefore use a baseline-aware rectification QA protocol as supporting infrastructure: it separates strict pass, pass-with-warning, and fail, and logs per-frame matching/stability diagnostics rather than relying on a single opaque gate.

Our main evaluation covers a 10-scene UAV-focused benchmark: 7 raw UAV captures and 3 aligned aerial multispectral scenes. The raw track validates end-to-end applicability (alignment, QA, reconstruction, products, and geometry diagnostics). The aligned track removes rectification confounds and supports fair common-mask comparison with MS-Splatting. Across this suite, UMGS-I and UMGS-J are near-tied and consistently preserve shared support. Scratch-MS and Unfrozen-support are useful upper-bound ablations for single-band fitting, but they do not satisfy the shared-support requirement for stable multispectral index synthesis.

The contributions are:

- A geometry-locked multispectral Gaussian formulation that uses RGB as the geometric carrier for stable spectral-index synthesis.

ACM XXXX-XXXX/2026/5-ART
https://doi.org/10.1145/nnnnnnn.nnnnnnn

- Two concrete UMGS realizations implementing the same lock with independent (UMGS-I) and joint (UMGS-J) spectral parameterizations.
- A UAV-focused benchmark protocol with image, index, depth-reference, cost, and external common-mask comparisons under unified reporting.
- Supporting raw-scene infrastructure for baseline-aware rectification QA with explicit diagnostic states and decision inputs.

2 Related Work

2.1 Radiance Fields and Gaussian Scene Representations

NeRF established neural volumetric rendering as a practical paradigm for novel view synthesis [Mildenhall et al. 2019, 2020]. Follow-up work improved anti-aliasing, unbounded scenes, and training speed, including mip-NeRF/mip-NeRF360, Instant-NGP, Plenoxels, TensoRF, and DVGO [Barron et al. 2022, 2021; Chen et al. 2022; Fridovich-Keil et al. 2022; M"uller et al. 2022; Sun et al. 2022]. Large-scene and explicit-factorized lines such as NSVF, K-Planes, Zip-NeRF, MERF, Mega-NeRF, and Block-NeRF further explored memory/scale trade-offs [Barron et al. 2023; Fridovich-Keil et al. 2023; Liu et al. 2020; Reiser et al. 2023; Tancik et al. 2022; Turki et al. 2022].

3D Gaussian Splatting (3DGS) moved from volumetric integration to explicit anisotropic splats, enabling faster optimization and real-time rendering [Kerbl et al. 2023]. Subsequent work improved alias control, structural priors, compactness, and scene scale, including Mip-Splatting, Scaffold-GS, 2DGS, SuGaR, LightGaussian, Compact3DGS, Dynamic 3DGS, CityGaussian, and CityGaussianV2 [Fan et al. 2024; Guédon and Lepetit 2024; Huang et al. 2024; Lee et al. 2024; Liu et al. 2024, 2025; Lu et al. 2024b; Luiten et al. 2024; Yu et al. 2024]. These methods target rendering quality, memory, or geometry fidelity in RGB-dominant settings. UMGS addresses a different constraint: preserving one stable 3D support while transferring low-texture UAV spectral appearance onto it.

2.2 Multispectral and Thermal Gaussian Reconstruction

Multispectral remote sensing has long emphasized calibration, cross-band consistency, and physically meaningful indices [Gitelson et al. 1996; Goetz et al. 1985; Huete 1988; Manolakis et al. 2016; Rouse et al. 1974]. In UAV workflows, geometric preprocessing and modality alignment are often the limiting factor for downstream reliability [Colomina and Molina 2014; Mathews and Jensen 2013; Turner et al. 2015; Zhang and Kovacs 2012]. While deep learning for spectral imagery is mature in classification/segmentation tasks [Audebert et al. 2019; Li et al. 2019; Paoletti et al. 2019; Zhong et al. 2018], 3D multispectral scene reconstruction is still emerging.

Recent Gaussian-based spectral works include MS-Splatting for multi-camera multispectral NVS, HyperGS for hyperspectral latent rendering, MSGS for wavelength-aware Gaussian modeling, and thermal-domain lines such as ThermalGaussian and Thermal3D-GS [Chen et al. 2024; Lu et al. 2024a; Meyer et al. 2025; Thirgood et al. 2025; Zhang et al. 2026]. These works confirm the feasibility of spectral Gaussian rendering and, in several cases, already store multiple spectral or modal signals in one Gaussian-based representation. MS-Splatting, for example, learns compact per-splat features and decodes spectral values with an MLP; HyperGS performs hyperspectral rendering in a learned latent space; ThermalGaussian learns registered RGB/thermal Gaussians after camera calibration. Our focus is complementary: raw UAV multispectral deployment, where spectral bands are weak geometry drivers and RGB/MS cameras can differ in FOV, resolution, and alignment. Rather than relying on each spectral band to define its own geometry, UMGS uses RGB as the geometric carrier and learns spectral appearance on the locked support.

2.3 SfM, Matching, and Cross-Modal Alignment

Practical reconstruction still depends on robust SfM/MVS foundations [Fischler and Bolles 1981; Hartley and Zisserman 2004; Sch"onberger and Frahm 2016; Sch"onberger et al. 2016]. Handcrafted and learned feature matching both remain relevant, from SIFT/SURF/ORB and ECC-style alignment to SuperPoint, SuperGlue, LoFTR, LightGlue, and RoMa [Bay et al. 2008; DeTone et al. 2018; Edstedt et al. 2024; Evangelidis and Psarakis 2008; Lindenberger et al. 2023; Lowe 2004; Rublee et al. 2011; Sarlin et al. 2020; Sun et al. 2021]. Robust estimators (LO-RANSAC, USAC, MAGSAC) are essential under modality gaps and outlier-heavy correspondences [Barath et al. 2019; Chum et al. 2003; Raghunandan et al. 2013].

Our raw-scene QA protocol follows this systems view: it reports match quality and geometric stability separately, preserves legacy strict criteria for strong baselines, and uses a selective warning path only for weak-baseline/high-modality-gap cases. This is a protocol contribution rather than a new matcher.

2.4 Depth and Geometry Diagnostics

Depth-based reconstruction metrics are widely used in multi-view stereo and neural reconstruction [Gu et al. 2020; Wang et al. 2021; Yao et al. 2018]. However, raw UAV multispectral captures usually lack globally calibrated metric scale. We therefore evaluate geometry by relative reference-depth agreement on held-out views, and report thresholded agreement curves rather than absolute-meter accuracy claims. This positions our geometry analysis as repeatability/self-consistency under controlled probes, aligned with multispectral product reliability objectives.

3 Method

3.1 Problem Formulation

Given RGB and multispectral UAV captures, our goal is to synthesize spectral novel views and spectral indices on a common 3D support. Let $\mathcal{G}$ denote Gaussian support parameters, including position, scale, rotation, opacity, and identity, and let $\mathcal{A}_b$ denote appearance parameters for band b. UMGS separates support estimation from spectral appearance learning: it estimates $\mathcal{G}$ once from RGB, then keeps $\mathcal{G}$ fixed while learning $\mathcal{A}_b$ for each multispectral band.

3.2 Stage 1: RGB Geometry Anchor

We first train an RGB Gaussian model with the 3DGS optimization pipeline [Kerbl et al. 2023]. RGB is used as the anchor because it provides stronger texture and more stable feature matching than the spectral bands in our UAV captures. The optimized support $\mathcal{G}^\star$ becomes the geometric carrier for all subsequent spectral rendering.

3.3 Stage 2: Geometry-Locked Spectral Appearance

UMGS-I (independent transfer). For each multispectral band b, we initialize from $\mathcal{G}^\star$, freeze all support parameters, and optimize only $\mathcal{A}_b$. This produces independent spectral appearance models whose Gaussians remain in one-to-one correspondence.

UMGS-J (joint band-bank transfer). We keep the same frozen $\mathcal{G}^\star$ but optimize band-specific appearance banks in one joint multispectral representation. Only the active bank is updated at each step, so all bands share the same support while retaining separate appearance capacity.

Geometry lock. Both variants enforce:

- fixed Gaussian count and identity;
- fixed xyz/scale/rotation/opacity;
- appearance-only parameter updates in stage 2.

These constraints define the shared support used by spectral indices. A variant that optimizes per-band xyz, scale, rotation, or opacity can still render each band independently, but its i-th Gaussian no longer denotes the same physical support across bands. Model-level ratios such as NDVI are then no longer defined on a consistent 3D support.

3.4 Raw-Scene Rectification and Baseline-Aware QA

Raw UAV scenes require spectral-to-RGB rectification before reconstruction. We estimate per-band homographies using deterministic candidate sampling and robust aggregation from feature matches. Because naive scaling baselines can be unreliable under cross-FOV and modality gaps, we use a baseline-aware QA protocol:

- **strict pass**: legacy edge+gradient improvement and geometry checks;
- **pass-with-warning**: allowed only for weak-baseline/high-modality-gap cases that satisfy match quality, stability, gradient gain, edge floor, and outlier-ratio constraints;
- **fail**: unstable geometry or insufficient matching evidence.

This protocol supports raw scenes only; aligned inputs use the same geometry-locked representation.

3.5 Products and Geometry Diagnostics

We render G/R/RE/NIR and compute NDVI, GNDVI, and NDRE under a common valid mask. Geometry is evaluated by relative depth agreement on held-out probe views:

$$\delta_{rel} = \frac{D_{model} - D_{ref}}{D_{ref}}.$$

Because global metric scale is not guaranteed in raw UAV reconstructions, we report relative-depth AbsRel and threshold agreements (1%, 5%, 10%) instead of meter-based absolute accuracy.

4 Experiments

4.1 Setup

Scenes. We evaluate the main paper on a 10-scene UAV-focused benchmark: 7 raw UAV scenes (`raw_self`, `raw001–raw006`) and 3 aligned aerial multispectral scenes (`ms_golf`, `ms_lake`, `ms_solar`). Raw scenes test rectification, QA, reconstruction, products, and geometry diagnostics. Aligned scenes remove rectification confounds and are used for external common-mask comparison.

Methods. We report two UMGS variants: UMGS-I, with independent per-band transfer, and UMGS-J, with joint band-bank transfer. MS-Splatting is the external baseline on aligned UAV-style scenes. Scratch-MS and Unfrozen-support are representative single-band upper-bound ablations; they are not product models because they do not preserve the shared Gaussian support required by model-level multispectral fusion.

Metrics. Image quality: PSNR/SSIM/LPIPS. Spectral means average G/R/RE/NIR. Index fidelity: MAE/RMSE/PSNR/SSIM for NDVI, GNDVI, NDRE. Geometry: relative-depth AbsRel and agreement at 1%, 5%, 10%. Cost: non-invasive GPU traces and exported Gaussian counts. Aligned external comparison uses a common-mask protocol; UMGS-I-shim and UMGS-J-shim are resolution-aligned comparison views only, not distinct trained variants.

4.2 Main Results on UAV Scenes

The main question is whether locking RGB geometry sacrifices spectral rendering quality. Table 1 shows that it does not: UMGS-I and UMGS-J remain near-identical on aggregate quality and geometry. On the 10-scene UAV-focused suite, both reach 0.843 spectral SSIM and 0.844 index SSIM; UMGS-I has a small edge on index RMSE (0.075 vs. 0.077). Depth aggregates are identical because both variants share the same locked support. We therefore use UMGS-I as the conservative primary line and treat UMGS-J as a joint-parameterization branch with comparable performance.

4.3 Index-Focused Analysis

Spectral-index synthesis is the downstream use case that motivates shared support. Table 2 breaks out NDVI, GNDVI, and NDRE. UMGS-I and UMGS-J are close on NDVI and GNDVI, while NDRE shows a larger RMSE/PSNR gap favoring UMGS-I. The joint-bank design is therefore viable but not consistently superior.

4.4 Aligned External Comparison

Aligned scenes remove the raw-rectification factor and isolate the reconstruction model. Under this common-mask setting, Table 3 shows a scene-dependent comparison with MS-Splatting: UMGS-J-shim is lower in LPIPS and has the same mean SSIM/SAM, while MS-Splatting is slightly higher in PSNR and slightly lower in spectral RMSE. Scene-level behavior in Table 4 is mixed, so the conclusion is not universal dominance but complementary behavior under a fair aligned protocol.

Raw unaligned data test a different question: whether an external method can be applied under its retained-subset path. Table 5 is therefore an applicability diagnostic rather than the main raw benchmark, because it does not represent the full raw protocol used by UMGS-I/UMGS-J.

4.5 Representative Ablations

The ablations test whether geometry locking is merely conservative or actually necessary. Table 6 compares the locked UMGS variants

Table 1. Main UMGS results over the 10-scene UAV-focused suite. RGB metrics for UMGS-J use the shared UMGS-I RGB anchor. Spectral metrics average G/R/RE/NIR. Index metrics average NDVI/GNDVI/NDRE. Depth uses the relative reference-depth protocol and averages G/R/RE/NIR.

Method	Split	RGB PSNR	RGB SSIM	MS PSNR	MS SSIM	MS LPIPS	Idx. RMSE	Idx. SSIM	AbsRel	Ag@5%	Ag@10%
UMGS-I	Raw 7	24.39	0.840	27.40	0.850	0.136	0.065	0.869	0.019	0.916	0.988
UMGS-I	Aligned 3	25.48	0.810	26.50	0.828	0.230	0.098	0.787	0.053	0.804	0.833
UMGS-I	UAV 10	24.72	0.831	27.13	0.843	0.164	0.075	0.844	0.029	0.883	0.942
UMGS-J	Raw 7	24.39	0.840	27.37	0.850	0.136	0.067	0.869	0.019	0.916	0.988
UMGS-J	Aligned 3	25.48	0.810	26.50	0.828	0.230	0.098	0.787	0.053	0.804	0.833
UMGS-J	UAV 10	24.72	0.831	27.11	0.843	0.164	0.077	0.844	0.029	0.883	0.942

Table 2. UAV-focused 10-scene spectral-index fidelity by vegetation index.

Method	Index	MAE	RMSE	PSNR	SSIM
UMGS-I	NDVI	0.058	0.083	28.49	0.847
UMGS-I	GNDVI	0.059	0.083	28.65	0.823
UMGS-I	NDRE	0.041	0.059	31.53	0.864
UMGS-J	NDVI	0.058	0.083	28.49	0.847
UMGS-J	GNDVI	0.059	0.083	28.66	0.823
UMGS-J	NDRE	0.043	0.063	31.31	0.863

Table 3. Aligned UAV-style common-mask comparison against MS-Splatting. Lower is better for LPIPS, 4-band RMSE, and SAM. UMGS shims are resolution-aligned comparison views, not separate training variants.

Method	PSNR	SSIM	LPIPS	4-band RMSE	SAM°
UMGS-I-shim	27.01	0.853	0.215	0.0545	5.38
UMGS-J-shim	27.01	0.853	0.215	0.0545	5.38
MS-Splatting	27.04	0.853	0.244	0.0541	5.38

with two upper-bound ablations on `raw_self`, `raw002`, and `ms_lake`. The unfrozen branch improves selected image metrics, but it is not a valid product model: on `ms_lake`, product construction fails the shared-support audit with an xyz mismatch of 6.84×10^{-2}. Scratch-MS diverges more strongly, including Gaussian-count mismatch across bands. These failures occur when multispectral products are assembled, showing that per-band quality gains alone are insufficient if support consistency is lost.

4.6 Cost

Finally, Table 7 reports coarse but traceable cost indicators on the UAV-focused runs. UMGS-J uses higher recorded peak VRAM but lower average recorded VRAM in the aggregated traces; both UMGS variants retain identical Gaussian counts.

5 Limitations

First, aligned external comparison is scene-dependent rather than one-sided. Some entries favor our geometry-locked shims, while others favor MS-Splatting. Our claim is therefore reliability and protocol applicability, not universal metric dominance.

Second, UMGS-I and UMGS-J are very close numerically. This validates that the geometry-lock principle is robust to independent and joint implementations. In the current paper, UMGS-J is best interpreted as a strong alternative branch rather than a strict replacement for UMGS-I.

Third, our geometry evaluation is a reference-depth consistency protocol, not dense metric ground truth. This is appropriate for cross-band support analysis under missing global scale, but it does not replace LiDAR-grade absolute geometry benchmarks.

Fourth, cost reporting is intentionally non-invasive and currently coarse. The reported VRAM/utilization/size statistics are traceable and reproducible, but a dedicated runtime microbenchmark (e.g., standardized per-view FPS across methods and resolutions) is still future work.

6 Conclusion

We presented UMGS, a geometry-locked multispectral 3D Gaussian Splatting framework for UAV spectral-index synthesis. The method anchors geometry in RGB, then learns spectral appearance with fixed support, so cross-band products are built on consistent Gaussian topology instead of loosely coupled per-band reconstructions.

Across the UAV-focused evaluation, UMGS-I and UMGS-J achieve near-identical quality and geometry behavior, showing that the geometry-lock principle is stable under both independent and joint spectral parameterizations. On aligned UAV-style scenes, comparison with MS-Splatting is competitive and scene-dependent, while on raw scenes the full pipeline demonstrates practical applicability with rectification QA, product synthesis, and depth-reference diagnostics.

The main takeaway is methodological: for multispectral UAV reconstruction, preserving shared support is a first-order constraint, not a secondary check after optimizing single-band image metrics.

References

Nicolas Audebert, Bertrand Le Saux, and Sébastien Lefèvre. 2019. Deep Learning for Classification of Hyperspectral Data: A Comparative Review. *IEEE Geoscience and Remote Sensing Magazine* 7, 2 (2019), 159–173.

Dániel Barath, Jiri Matas, and Jana Noskova. 2019. MAGSAC: Marginalizing Sample Consensus. In *CVPR*.

Jonathan T. Barron, Ben Mildenhall, Matthew Tancik, Peter Hedman, Ricardo Martin-Brualla, and Pratul P. Srinivasan. 2022. Mip-NeRF 360: Unbounded Anti-Aliased Neural Radiance Fields. In *CVPR*.

Jonathan T. Barron, Ben Mildenhall, Dor Verbin, Peter Hedman, and Pratul P. Srinivasan. 2023. Zip-NeRF: Anti-Aliased Grid-Based Neural Radiance Fields. *ICCV* (2023).

Jonathan T. Barron, Ben Mildenhall, Dor Verbin, Pratul P. Srinivasan, and Peter Hedman. 2021. Mip-NeRF: A Multiscale Representation for Anti-Aliasing Neural Radiance Fields. In *ICCV*.

Herbert Bay, Andreas Ess, Tinne Tuytelaars, and Luc Van Gool. 2008. Speeded-Up Robust Features (SURF). *Computer Vision and Image Understanding* 110, 3 (2008), 346–359.

Anpei Chen, Zexiang Xu, Andreas Geiger, Jingyi Yu, and Hao Su. 2022. TensoRF: Tensorial Radiance Fields. In *ECCV*.

Qian Chen, Shihao Shu, and Xiangzhi Bai. 2024. Thermal3D-GS: Physics-induced 3D Gaussians for Thermal Infrared Novel-view Synthesis. arXiv:2409.08042 [cs.CV] https://arxiv.org/abs/2409.08042

Ondrej Chum, Jiri Matas, and Josef Kittler. 2003. Locally Optimized RANSAC. In *DAGM*.

Table 4. Scene-level aligned UAV-style common-mask comparison for UMGS-I-shim and MS-Splatting. PSNR/SSIM/LPIPS average G/R/RE/NIR; 4-band RMSE and SAM evaluate the spectral vector.

Scene	UMGS PSNR	MMS PSNR	UMGS SSIM	MMS SSIM	UMGS LPIPS	MMS LPIPS	UMGS RMSE	MMS RMSE	UMGS SAM	MMS SAM
ms_golf	31.33	32.12	0.928	0.935	0.176	0.234	0.0300	0.0273	2.77	3.11
ms_lake	22.02	22.79	0.740	0.731	0.350	0.374	0.0887	0.0817	8.84	7.88
ms_solar	27.72	26.21	0.893	0.893	0.117	0.123	0.0448	0.0532	4.52	5.16

Table 5. Raw `self_m3m` retained-subset diagnostic. This is an applicability check for the external pipeline under its retained subset, not the main raw full-scene comparison.

Method	PSNR	SSIM	LPIPS	4-band RMSE	SAM°
UMGS-I-retained	28.35	0.834	0.203	0.0502	3.98
UMGS-J-retained	28.33	0.834	0.204	0.0503	4.00
MS-Splatting-default	31.07	0.933	0.107	0.0379	3.95

Ismael Colomina and Pere Molina. 2014. Unmanned Aerial Systems for Photogrammetry and Remote Sensing: A Review. *ISPRS Journal of Photogrammetry and Remote Sensing* 92 (2014), 79–97.

Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. 2018. SuperPoint: Self-Supervised Interest Point Detection and Description. In *CVPR Workshops*.

Johan Edstedt, Qiyuan Wang, Victor Larsson, Daniel Barath, and Torsten Sattler. 2024. RoMa: Robust Dense Feature Matching. In *CVPR*.

Georgios D. Evangelidis and Emmanouil Z. Psarakis. 2008. Parametric Image Alignment Using Enhanced Correlation Coefficient Maximization. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 30, 10 (2008), 1858–1865.

Zhiwen Fan, Wenqi Xian, Jiaxin Yang, Dejia Xu, Deva Ramanan, Jan Kautz, and Zhe Lin. 2024. LightGaussian: Unbounded 3D Gaussian Compression with 15x Reduction and 200+ FPS. *NeurIPS* (2024).

Martin A. Fischler and Robert C. Bolles. 1981. Random Sample Consensus: A Paradigm for Model Fitting with Applications to Image Analysis and Automated Cartography. *Commun. ACM* 24, 6 (1981), 381–395.

Sara Fridovich-Keil, Giacomo Meanti, Francesco Poiesi, Michael Rubinstein, and Angjoo Kanazawa. 2023. K-Planes: Explicit Radiance Fields in Space, Time, and Appearance. In *CVPR*.

Sara Fridovich-Keil, Alex Yu, Matthew Tancik, Qinhong Jiang, Benjamin Recht, and Angjoo Kanazawa. 2022. Plenoxels: Radiance Fields without Neural Networks. In *CVPR*.

Anatoly A. Gitelson, Yosef J. Kaufman, and Mark N. Merzlyak. 1996. Use of a Green Channel in Remote Sensing of Global Vegetation from EOS-MODIS. *Remote Sensing of Environment* 58, 3 (1996), 289–298.

Alexander F. H. Goetz, George Vane, John E. Solomon, and Barry N. Rock. 1985. Imaging Spectrometry for Earth Remote Sensing. *Science* 228, 4704 (1985), 1147–1153.

Xiaodong Gu, Zhiqiang Fan, Siyu Zhu, Zhaoyang Dai, Feitong Tan, and Ping Tan. 2020. CAS-MVSNet: Cascade Cost Volume for High-Resolution Multi-View Stereo and Stereo Matching. In *CVPR*.

Antoine Guédon and Vincent Lepetit. 2024. SuGaR: Surface-Aligned Gaussian Splatting for Efficient 3D Mesh Reconstruction and High-Quality Novel View Synthesis. *CVPR* (2024).

Richard Hartley and Andrew Zisserman. 2004. *Multiple View Geometry in Computer Vision* (2nd ed.). Cambridge University Press.

Binbin Huang, Zehao Yu, Anpei Chen, Andreas Geiger, Shenghua Gao, and Hao Su. 2024. 2D Gaussian Splatting for Geometrically Accurate Radiance Fields. *SIGGRAPH* (2024).

Alfredo R. Huete. 1988. A Soil-Adjusted Vegetation Index (SAVI). *Remote Sensing of Environment* 25, 3 (1988), 295–309.

Bernhard Kerbl, Georgios Kopanas, Thomas Leimk"uhler, and George Drettakis. 2023. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics* 42, 4 (2023), 139:1–139:14.

Jaesung Lee, Jihoon Kim, Jaewoo Park, and Minsik Lee. 2024. Compact3DGS: Memory-Efficient 3D Gaussian Splatting. *arXiv preprint arXiv:2403.18159* (2024).

Wen Li, Geng Sun, Yunsong Li, and Qian Du. 2019. Deep Learning for Hyperspectral Image Classification: An Overview. *IEEE Transactions on Geoscience and Remote Sensing* 57, 9 (2019), 6690–6709.

Philipp Lindenberger, Paul-Edouard Sarlin, and Marc Pollefeys. 2023. LightGlue: Local Feature Matching at Light Speed. In *ICCV*.

Lingjie Liu, Jiatao Gu, Kyaw Zaw Lin, Tat-Seng Chua, and Christian Theobalt. 2020. Neural Sparse Voxel Fields. *NeurIPS* (2020).

Yang Liu, He Guan, Chuanchen Luo, Lue Fan, Naiyan Wang, Junran Peng, and Zhaoxiang Zhang. 2024. CityGaussian: Real-Time High-Quality Large-Scale Scene Rendering with Gaussians. *ECCV* (2024).

Yang Liu, Chuanchen Luo, Zhongkai Mao, Junran Peng, and Zhaoxiang Zhang. 2025. CityGaussianV2: Efficient and Geometrically Accurate Reconstruction for Large-Scale Scenes. In *ICLR*.

David G. Lowe. 2004. Distinctive Image Features from Scale-Invariant Keypoints. *International Journal of Computer Vision* 60, 2 (2004), 91–110.

Jiahao Lu, Xiangli Yao, Yinda Zhang, Yao Yao, Hai Li, Tian Fang, and Long Quan. 2024b. Scaffold-GS: Structured 3D Gaussians for View-Adaptive Rendering. *CVPR* (2024).

Rongfeng Lu, Hangyu Chen, Zunjie Zhu, Yuhang Qin, Ming Lu, Le Zhang, Chenggang Yan, and Anke Xue. 2024a. ThermalGaussian: Thermal 3D Gaussian Splatting. arXiv:2409.07200 [cs.CV] https://arxiv.org/abs/2409.07200

Jonathon Luiten, Leonidas Guibas, Deva Ramanan, and Florian Bernard. 2024. Dynamic 3D Gaussians: Tracking by Persistent Dynamic View Synthesis. *3DV* (2024).

Dimitris Manolakis, Ronald Lockwood, Thomas Cooley, and John Jacobson. 2016. *Hyperspectral Imaging Remote Sensing: Physics, Sensors, and Algorithms.* Cambridge University Press.

A. J. Mathews and S. Jensen. 2013. Visualizing and Quantifying Vineyard Canopy LAI Using an Unmanned Aerial Vehicle (UAV) Collected High Density Structure from Motion Point Cloud. *Remote Sensing* 5, 5 (2013), 2164–2183.

Lukas Meyer, Josef Grün, Maximilian Weiherer, Bernhard Egger, Marc Stamminger, and Linus Franke. 2025. Multi-Spectral Gaussian Splatting with Neural Color Representation. arXiv:2506.03407 [cs.CV] https://arxiv.org/abs/2506.03407

Ben Mildenhall, Pratul P. Srinivasan, Rodrigo Ortiz-Cayon, Nima Khademi Kalantari, and Ren Ng. 2019. Local Light Field Fusion: Practical View Synthesis with Prescriptive Sampling Guidelines. In *ACM Transactions on Graphics*.

Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. 2020. NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis. In *ECCV*.

Thomas M"uller, Alex Evans, Christoph Schied, and Alexander Keller. 2022. Instant Neural Graphics Primitives with a Multiresolution Hash Encoding. *ACM Transactions on Graphics* 41, 4 (2022).

M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza. 2019. Deep Learning Classifiers for Hyperspectral Imaging: A Review. *ISPRS Journal of Photogrammetry and Remote Sensing* 158 (2019), 279–317.

Raguram Raghunandan, Ondrej Chum, Marc Pollefeys, Jiri Matas, and Jan-Michael Frahm. 2013. USAC: A Universal Framework for Random Sample Consensus. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 35, 8 (2013), 2022–2038.

Christian Reiser, Songyou Peng, Yuanqing Zhang, and Andreas Geiger. 2023. MERF: Memory-Efficient Radiance Fields for Real-Time View Synthesis in Unbounded Scenes. *SIGGRAPH Asia Conference Papers* (2023).

John W. Rouse, Robert H. Haas, John A. Schell, and Donald W. Deering. 1974. Monitoring Vegetation Systems in the Great Plains with ERTS. In *Third Earth Resources Technology Satellite Symposium*.

Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski. 2011. ORB: An Efficient Alternative to SIFT or SURF. In *ICCV*.

Paul-Edouard Sarlin, Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. 2020. SuperGlue: Learning Feature Matching with Graph Neural Networks. In *CVPR*.

Johannes L. Sch"onberger and Jan-Michael Frahm. 2016. Structure-from-Motion Revisited. In *CVPR*.

Johannes L. Sch"onberger, Enliang Zheng, Marc Pollefeys, and Jan-Michael Frahm. 2016. Pixelwise View Selection for Unstructured Multi-View Stereo. In *ECCV*.

Cheng Sun, Min Sun, and Hwann-Tzong Chen. 2022. Direct Voxel Grid Optimization: Super-fast Convergence for Radiance Fields Reconstruction. In *CVPR*.

Jiaming Sun, Zixin Shen, Yuangdong Xu, Zerun Wang, Qi Bao, Xiaowei Zhou, Yanning Zhang, and Junsong Yuan. 2021. LoFTR: Detector-Free Local Feature Matching with Transformers. In *CVPR*.

Matthew Tancik, Vincent Casser, Xinchen Yan, Sabeek Pradhan, Ben Mildenhall, Pieter Abbeel, Ren Ng, and Angjoo Kanazawa. 2022. Block-NeRF: Scalable Large Scene Neural View Synthesis. *CVPR* (2022).

Christopher Thirgood, Oscar Mendez, Erin Chao Ling, Jon Storey, and Simon Hadfield. 2025. HyperGS: Hyperspectral 3D Gaussian Splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 5970–5979.

Table 6. Representative 3-scene ablation summary on `raw_self`, `raw002`, and `ms_lake`. Spectral metrics average G/R/RE/NIR; index metrics average NDVI/GNDVI/NDRE. Scratch-MS and Unfrozen-support are single-band upper-bound ablations, not valid product models, because their shared-support audits fail.

Method	MS PSNR	MS SSIM	MS LPIPS	Idx. RMSE	AbsRel	Ag@5%	Shared	Products
UMGS-I	25.35	0.786	0.228	0.100	0.045	0.833	yes	yes
UMGS-J	25.33	0.786	0.228	0.100	0.045	0.833	yes	yes
Scratch-MS	25.88	0.780	0.289	0.116	0.035	0.560	no	invalid
Unfrozen-support	26.27	0.794	0.218	0.105	0.046	0.830	no	invalid

Table 7. Coarse recorded cost indicators on the UAV-focused suite. Values are aggregated from non-invasive GPU traces and exported PLY vertex counts; they are intended as a first cost table, not a microbenchmark.

Method	Peak VRAM GB	Avg. VRAM GB	Peak util. %	Gaussians M
UMGS-I	27.45	3.99	99.0	4.31
UMGS-J	35.04	1.50	99.0	4.31

Hadi Turki, Deva Ramanan, and Mahadev Satyanarayanan. 2022. Mega-NeRF: Scalable Construction of Large-Scale NeRFs for Virtual Fly-Throughs. In *CVPR*.

Darren Turner, Andreas Lucieer, and Simon M. de Jong. 2015. Time Series Analysis of Landslide Dynamics Using an Unmanned Aerial Vehicle (UAV). *Remote Sensing* 7, 2 (2015), 1736–1757.

Fangjinhua Wang, Silvano Galliani, Christoph Vogel, Marc Pollefeys, and Jan Kautz. 2021. PatchmatchNet: Learned Multi-View Patchmatch Stereo. In *CVPR*.

Yao Yao, Zixin Luo, Shiwei Li, Tian Fang, and Long Quan. 2018. MVSNet: Depth Inference for Unstructured Multi-view Stereo. In *ECCV*.

Zehao Yu, Anpei Chen, Andreas Geiger, Shenghua Gao, and Hao Su. 2024. Mip-Splatting: Alias-Free 3D Gaussian Splatting. *CVPR* (2024).

Chao Zhang and John M. Kovacs. 2012. The Application of Small Unmanned Aerial Systems for Precision Agriculture: A Review. *Precision Agriculture* 13, 6 (2012), 693–712.

Fanglue Zhang et al. 2026. MSGS: Multispectral 3D Gaussian Splatting. arXiv:2604.13340 [cs.CV] https://arxiv.org/abs/2604.13340

Zilong Zhong, Jonathan Li, Ziyu Luo, and Michael Chapman. 2018. Spectral-Spatial Residual Network for Hyperspectral Image Classification: A 3-D Deep Learning Framework. *IEEE Transactions on Geoscience and Remote Sensing* 56, 2 (2018), 847–858.